

**ROBOFEST- GUJARAT 6.0**

**Guidelines for submission of Ideation Proposal**  
**Senior category**

All teams must be present and ready to perform their challenge. The committee may assign any team to the task at random. Any kind of absence and lack of readiness will be subject to suitable actions, including the deduction of score/disqualification.

Students are required to perform their task and answer the questions. Mentors should not be involved in the task. The committee may allow mentors during question-answer if it feels necessary, depending on the situation.

Sample images given in these guidelines are only for visual understanding.

**Intelligent Ground Vehicle Competition (IGVC)**

**Game Description:**

The Intelligent Ground Vehicle Competition (IGVC) challenges participants to develop an autonomous ground robot capable of navigating complex terrain and obstacle courses without human intervention. The robot must demonstrate localization, path planning, obstacle avoidance, and decision-making skills using only onboard sensors and computing.

This simulation mirrors real-world applications of autonomous navigation in fields like agriculture, defence, space exploration, and search & rescue.

**Game Field Layout:**

- Arena Size: Minimum 12m x 8m rectangular track area.
- Surface: Outdoor surface with variable friction zones (e.g., carpet, tiles, ramps, uneven surface, natural grass etc.) The arena will neither have any track boundary nor any guiding line to follow.
- Obstacle Zones:
  - Includes randomly-placed static and dynamic obstacles.
  - Includes Arches, underpass, tunnel, cones, ramps (max 40-degree incline), dual incline (both sides) and narrow passages. The exact dimensions of these way-zones may vary depending on the width of the robot.
- Checkpoints: 4-checkpoints at the end of each stage.
- Start/Finish Zone: Marked areas at opposite corners of the arena.
- Navigation Aids: May include QR codes at checkpoints.

**Challenge Tasks:**

1. Stage 1 – Basic Navigation (5 Points):
  - The Robot must autonomously follow a path for 3 minutes.
  - Simple left and right turns included.
2. Stage 2 – Obstacle Avoidance (15 Points):
  - Encounter static obstacles such as cones and boxes.
  - The Robot must identify and detour without crossing the arena boundaries.
3. Stage 3 – Terrain Challenge (25 Points):
  - Includes uneven surfaces, small ramps, underpass, tunnel and a narrow bridge.
  - Requires stability and path adjustment.
  - Able to grasp an object en-route
4. Stage 4 – Smart Decision Zone (20 Points):
  - Multiple paths, only one is correct.

- The Robot must scan QR/AR code or symbolic signboard to select correct route.
- 5. Stage 5 – Dynamic Obstacle Detection (20 Points):
  - A slow-moving object (e.g., pendulum or rolling ball or small robot).
  - The Robot must stop, wait, and resume after the path clears.
- 6. Bonus Task—Reinforcement Learning (RL)-induced Navigation (15 points)
  - Expert navigation by the robot upon investigation of previous similar outdoor scenario with grasping OR
  - Efficient navigation of the robot under feature-deficient / degraded environment outdoor with grasping

The dimensions / size of the vehicle can be decided by the team.

The robot should be able to demonstrate collision-free navigation as well as grasping an object under a cluttered field.

Approximate size of the underpass: 10 ft x 4ft x 8 ft. (Length x Breadth x Depth). In other words, the robot has to move smoothly over a ramp beneath the horizontal surface upto a depth of 8 ft. (approx.). Approximate size of the narrow passage: 10 ft. x 4 ft (planar). Approximate dimensions of the tunnel: 15 ft x 4 ft x 5 ft (Length x Breadth x Height). However, pl. note that these are approximate dimensions that may vary slightly based on the availability of site and other logistic support of ROBOFEST authorities.

## **Stage 1: Proof of Concept (PoC)**

### **Objective**

To evaluate the fundamental capabilities of the Intelligent Ground Vehicle including:

- Autonomous navigation
- Sensor integration
- Obstacle detection
- Basic autonomous decision making

### **PoC Challenge Tasks**

#### **1. Autonomous Straight Navigation**

The IGV must:

- Move autonomously for a predefined distance and/or time-period.
- Stop accurately at the designated point.

#### **2. Terrain Challenge:**

- Navigate a ramp
- Navigate uneven surfaces
- Complete the terrain without manual intervention.

#### **3. Obstacle Detection & Avoidance**

The vehicle shall:

- Detect a stationary obstacle.
- Safely avoid the obstacle.
- Resume the original path.

#### **4. Traffic Signal Recognition**

The vehicle shall identify any one of the following:

- Stop Sign

- Left Turn
- Right Turn
- Speed Limit
- Traffic Signal

and execute the corresponding action.

## 5. Controlled Parking Demonstration

The vehicle shall:

- Detect a designated parking zone.
- Park within the marked area.
- Stop completely.

## Technical Constraints (PoC Stage)

- Fully autonomous operation is preferred.
- Semi-autonomous mode may be permitted for safety.
- No restriction on vehicle dimensions (as per Robofest 5.0 guidelines).
- Any onboard processor may be used.
- Teams may use vision, LiDAR, ultrasonic, IR or equivalent sensors.

## Evaluation Criteria (PoC)

| Criteria                             | Marks |
|--------------------------------------|-------|
| Navigation Control                   | 20    |
| Track Following♣ / Terrain Challenge | 20    |
| Obstacle Detection & Avoidance       | 20    |
| Traffic Sign Recognition             | 15    |
| Parking Accuracy                     | 15    |
| System Integration                   | 5     |
| Innovation                           | 5     |

♣: Makeshift 'Track' may be provided by GUJCOST on case-to-case basis.

## Stage 2: Grand Finale (Mission Execution)

### Objective

To demonstrate a fully autonomous Intelligent Ground Vehicle capable of navigating a complex real-world environment using multiple sensors and intelligent decision-making.

### Challenge Description

#### Mission: Autonomous Urban Mobility Challenge

The IGV must:

- Navigate through an urban-style arena.
- Follow traffic regulations.
- Detect and avoid dynamic obstacles.
- Perform intelligent route selection.
- Complete the mission safely within the specified time.

### Advanced Challenge Tasks

## **1. Autonomous Navigation**

The vehicle shall:

- Navigate underpass, ramp and narrow bridge
- Negotiate sharp curves.
- Cross intersections.

## **2. Intelligent Traffic Management**

The vehicle shall:

- Detect traffic lights.
- Recognize traffic signs.
- Respond appropriately.

## **3. Dynamic Obstacle Avoidance**

The vehicle shall:

- Detect moving obstacles.
- Stop when necessary.
- Re-plan the path.
- Continue safely.

## **4. Smart Decision Zone**

The vehicle shall:

- Reach an intersection.
- Scan QR code / AprilTag/ symbolic signboard.
- Select the correct route autonomously.

## **5. Grasp an Object**

The vehicle shall:

- Approach a static object en-route
- Grasp the object with adequate force & form closure (without slippage).
- Proceed further in the arena with the grasped object till the end

## **6. Precision Parking Challenge**

The vehicle shall:

- Detect the assigned parking slot.
- Park completely within the designated boundary.
- Avoid contact with surrounding objects.

## **7. Emergency Response Zone (Advanced)**

The vehicle shall:

- Detect a blocked route.
- Select an alternate path.
- Reach the destination autonomously.

## **8. Return Mission**

After completing all assigned tasks, the vehicle shall:

- Return to the designated finish zone.
- Stop accurately within the marked area.

## Arena Design

The arena may include:

- Multiple road intersections
- Traffic signals
- Zebra crossings
- Lane markings
- Tunnel/underpass
- Narrow passage
- Speed breaker
- Inclined ramp
- QR/AprilTag decision zones
- Static and dynamic obstacles

Arena dimensions shall remain broadly similar to Robofest Gujarat 5.0 with additional modular challenge zones.

## Technical Requirements

- Fully autonomous operation.
- Onboard processing mandatory.
- Multiple sensor integration encouraged:
  - Camera
  - LiDAR
  - Ultrasonic Sensors
  - IMU
  - GPS (optional)
  - Wheel Encoders
- Emergency stop mechanism shall be mandatory.

## Evaluation Criteria (Grand Finale)

| Criteria                                  | Marks |
|-------------------------------------------|-------|
| Autonomous Navigation                     | 20    |
| Traffic Rule Compliance                   | 15    |
| Obstacle Avoidance and emergency response | 20    |
| Smart Decision Making                     | 10    |
| Precision Parking                         | 10    |
| Grasping of Object                        | 10    |
| Mission Completion                        | 10    |
| Robustness & Reliability                  | 5     |

## Scoring Logic

- Successful completion of each mission: **Awarded as per evaluation criteria**
- Time-efficient completion: **Bonus +5**

## Penalties

- Collision with obstacle: **-5**
- Traffic signal violation: **-5**
- Manual intervention: **-10**

## Design Expectations:

- Completely autonomous operation

- Sensors: IR/Ultrasonic, LIDAR, or vision-based navigation
- Onboard computing (Raspberry Pi, Jetson Nano, etc.)
- Fail-safe mechanism (manual kill switch)

Only 1 re-run permitted with due deduction of penalty.

#### **Tie Breaker**

1. Maximum challenges completed.
2. Minimum penalty points.
3. Fastest completion time.
4. Decision of the Technical Advisory Committee (TAC) shall be final.

**Note:** The guidelines related to the eligibility, timelines, formats of submissions, prizes and conduct are published on the Robofest Gujarat Webportal and shall be binding to all participating teams.